

CATCH: Counterfactual Anatomical Tissue Inpainting with Conditional Haar-Wavelet Diffusion

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Abstract. The BraTS local-synthesis task asks for plausible tumor-free tissue to be generated inside a masked region of a T1-weighted brain MRI while preserving the observed anatomy. We present CATCH, a conditional 3D diffusion framework operating on an invertible Haar-wavelet representation. Its denoiser receives noisy target subbands, voided-image subbands, and a signed mask. A tumor-excluded wavelet loss with a hole-localized term provides supervision, while hard compositing preserves every observed voxel. We compare fixed-mask training with strict random tumor-component augmentation and a configured weighted mixture of tumor-derived, blob, and ellipsoid masks. Arm-specific checkpoints and trajectory policies are selected on development data, then frozen for a held-out 75-case confirmation restricted to artificially removed healthy tissue. The selected weighted-mixture pipeline (five trajectories, voxel-wise mean) achieved SSIM 0.7972 ± 0.1308 , PSNR 19.18 ± 1.80 dB, and MSE 0.009874 ± 0.005359 (mean $\pm$ sample SD). Against compute-matched random augmentation, its mean improvements were 0.0188 SSIM (95% bootstrap CI: 0.0127–0.0249), 0.948 dB PSNR, and 0.002641 MSE reduction; all three corrected paired tests favored weighted mixture. Fixed-mask single-trajectory inference achieved SSIM 0.7561 ± 0.1615 . These within-CATCH results support the complete weighted-mixture policy, but one training seed, arm-specific checkpoint selection, and unequal trajectory counts relative to fixed mask limit causal and cross-method conclusions.

Keywords: MRI inpainting · Brain MRI · Medical image synthesis · Diffusion models · Wavelet transform · BraTS

1 Introduction

Brain tumors alter local anatomy and violate the healthy-tissue assumptions of many registration, parcellation, and atlas-based neuroimaging pipelines. Counterfactual inpainting replaces a pathological region with plausible tumor-free

tissue so that such tools can operate on a completed image. The BraTS local-synthesis task [10] formalizes this problem: given a voided T1n (*T1-weighted native*) volume and its inpainting mask, synthesize the missing anatomy while leaving all observed voxels unchanged.

Diffusion models [9] provide effective conditional generators [12,16]. Direct full-resolution 3D diffusion is memory intensive, however. Wavelet diffusion models (WDM) [8] reduce this cost through a fixed, invertible spatial transform, and conditional WDM [7] extends the formulation to image-to-image tasks. We adapt it as CATCH (**C**ounterfactual **A**natomical **T**issue Inpainting with **C**onditional **H**aar-wavelet diffusion) and ask how healthy training holes should be sampled.

Our contributions are:

- conditional 3D Haar-wavelet diffusion with tumor-excluded supervision and voxel-exact preservation outside the requested hole;
- a within-CATCH comparison of fixed-mask, strict random, and weighted-mixture healthy-mask training variants, with compute-matched random versus weighted inference;
- development-only checkpoint and trajectory-policy selection followed by a frozen 75-case confirmation with paired uncertainty, patient-disjoint sensitivity, and audited examples.

2 Related Work

Diffusion and inpainting. Denoising diffusion probabilistic models (DDPMs) learn the reverse of a gradual noising process [9]; score-based formulations connect this process to stochastic differential equations [18]. RePaint conditions an unconditional model by repeatedly restoring known pixels during sampling [12], whereas Palette passes the observed image to a conditional denoiser [16]. CATCH follows the latter approach and enforces the known region once more by final hard compositing.

Volumetric medical synthesis. Learned latent spaces reduce high-resolution diffusion cost [14]; WDM instead uses an exactly invertible 3D wavelet transform [8]. Its conditional extension supports volumetric image-to-image synthesis [7]. Diffusion has also been applied directly to 3D healthy-brain inpainting [6] in the BraTS local-synthesis setting [10], whose data build on the wider BraTS benchmark [13,4,1]. More recently, fastWDM3D targets the substantial sampling cost of wavelet-space healthy-tissue inpainting with a shortened reverse process [5]. Our work retains full ancestral sampling and tailors mask representation, loss support, online hole sampling, and development-selected trajectory aggregation; differing cohorts and protocols preclude a direct score comparison.

3 Method

CATCH adapts Tooth-Diffusion [17] from dental CBCT synthesis to brain-MRI inpainting.

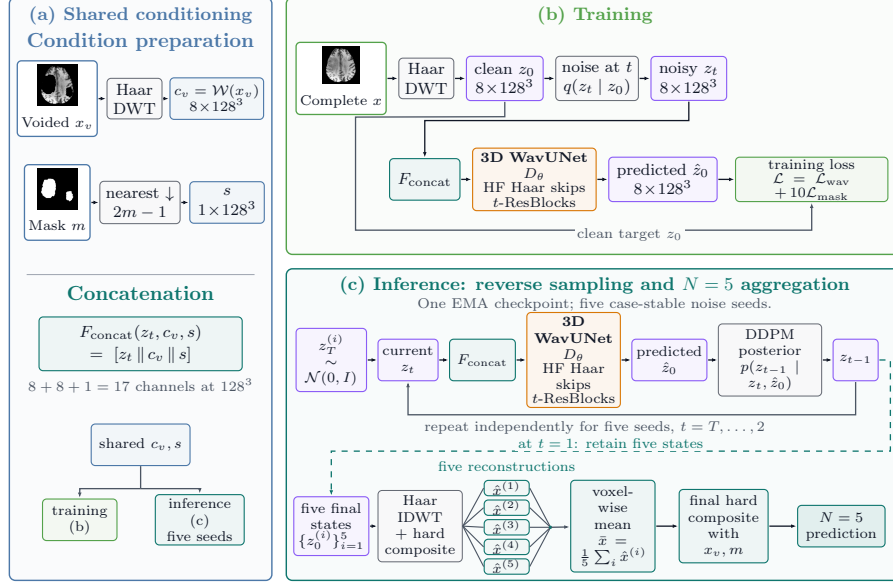


Fig. 1. CATCH training and inference. Haar transforms produce eight subbands; noisy target, voided-input, and signed-mask channels form the 17-channel WavUNet input. Predictions are inverse-transformed and hard-composited. Random and weighted models average five case-stable trajectories from one EMA checkpoint; fixed mask uses $N = 1$. Thumbnails were selected without reconstruction metrics.

Let x_v be the voided T1n volume and $m \in \{0, 1\}^{H \times W \times D}$ the full region to inpaint. During training, the provided mask is partitioned into an artificially removed healthy region m_h and a real tumor region m_u . The complete T1n volume x supplies a target in m_h but still contains pathology in m_u , so the latter is excluded from supervision. Only x_v and m are available to the model at inference.

3.1 Conditional Wavelet Diffusion

Volumes are reoriented to RAS and center-padded from $240 \times 240 \times 155$ to 256^3 without resampling. A one-level 3D Haar transform $\mathcal{W}$ produces eight 128^3 subbands; the LLL subband is divided by three during modeling and restored before inversion, following WDM [8]. Thus the transform is fixed and invertible rather than a learned autoencoder.

With $z_0 = \mathcal{W}(x)$, the $T = 1000$ forward process is

$$q(z_t | z_0) = \mathcal{N}(z_t; \sqrt{\bar{\alpha}_t} z_0, (1 - \bar{\alpha}_t) \mathbf{I}), \quad (1)$$

using the variance-preserving SDE schedule [18] with $\beta_{\min} = 0.1$ and $\beta_{\max} = 20$. The network predicts the clean coefficients $\hat{z}_0$ directly and uses fixed-large reverse variance.

The voided image becomes $c_v = \mathcal{W}(x_v)$. We nearest-neighbor downsample m to the wavelet grid and encode it as $s = 2\text{NN}(m) - 1 \in \{-1, 1\}^{128^3}$. The denoiser receives

$$\hat{z}_0 = D_\theta([z_t \| c_v \| s], t), \quad (2)$$

a 17-channel input. We call D_θ *WavUNet*, a fully 3D wavelet U-Net [15,8] with base width 64, channel multipliers (1, 2, 2, 4, 4, 4), two residual blocks per level, GroupNorm, SiLU, timestep scale-and-shift modulation, wavelet skip connections, and no self-attention.

3.2 Healthy-Tissue Objective

The target and condition share the inference-available scale $a = \max(x_v)$, are clipped to $[0, 1]$, and mapped to $[-1, 1]$. Let $e_{k,p} = (\hat{z}_0^{(k)}(p) - z_0^{(k)}(p))^2$. A wavelet cell is assigned $u_p = 0$ if its corresponding 2^3 voxel cell intersects m_u , and $u_p = 1$ otherwise. The base loss is

$$\mathcal{L}_{\text{wav}} = \frac{1}{8|\Omega|} \sum_{k=1}^8 \sum_{p \in \Omega} u_p e_{k,p}. \quad (3)$$

For the hole-focused term, cells intersecting m_h form a binary coverage map that is Gaussian-smoothed ($\sigma = 2$ on the wavelet grid) and multiplied by u to produce $w_p \geq 0$. We use

$$\mathcal{L}_{\text{mask}} = \frac{\sum_{k,p} w_p e_{k,p}}{8 \sum_p w_p}, \quad \mathcal{L} = \mathcal{L}_{\text{wav}} + 10\mathcal{L}_{\text{mask}}. \quad (4)$$

Normalizing by active mask mass keeps the regional coefficient comparable across hole sizes. Both terms exclude cells touching the real tumor.

3.3 Healthy-Mask Training Variants

The *fixed-mask* variant uses the supplied healthy mask. Online variants sample m'_h , void $m'_h \cup m_u$, and supervise m'_h while excluding m_u . Their tumor-component bank contains only the 1,151 optimization cases, and each variant has five repeat slots per case.

Random augmentation is a strict reference-style policy: tumor components only; current-case components allowed; inverse tumor-to-brain-ratio rank sampling within a conditioned 10% window; reference flips and rotations; five slots fixed across epochs; and an error after exhausted placement.

Weighted-mixture augmentation configures 80% tumor components, 15% irregular blobs, and 5% ellipsoids. It excludes current-case components, samples empirical volumes, applies bounded 3D rotations and optional anisotropic scaling, and refreshes slots by epoch. Generated masks are connected, contain 800–225,070 voxels, remain at least five voxels from m_u , and overlap background by at most 25%. Exhausted generation falls back to a supplied mask verified only as nonempty and nonoverlapping with m_u .

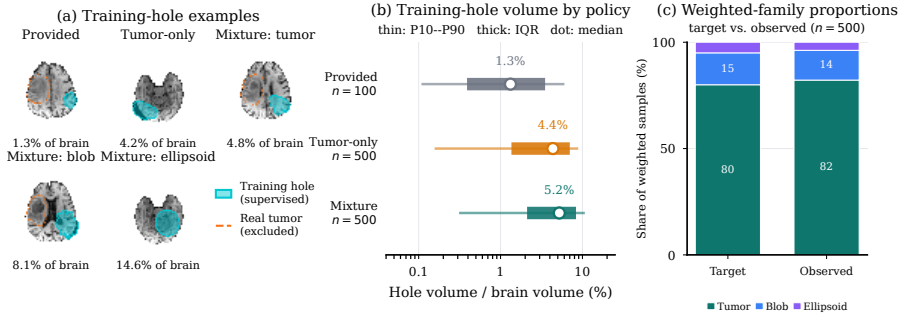


Fig. 2. Training-mask audit. (a) Supplied and generated examples; cyan marks supervised voxels, dashed orange the excluded tumor, and labels give 3D hole/brain volume. (b) Epoch-0 volume distributions by policy (P10–P90, interquartile range, and median; $n = 100/500/500$). (c) Target and observed weighted-family shares: tumor/blob/ellipsoid 80/15/5% and 82.2/14.0/3.8% (411/70/19); no fallback.

3.4 Sampling and Compositing

We use exponential-moving-average (EMA) weights and the complete 1000-step ancestral reverse process. At each step, `clip_denoised=True` inverse-transforms the predicted clean state, clips it in normalized voxel space, and transforms it back. The final reconstruction is restored to native intensity, shape, and orientation, then hard-composited:

$$\hat{x} = x_v \odot (1 - m) + \hat{x}_{\text{gen}} \odot m, \quad (5)$$

Global seed 0 and the case- and member-stable `brats-validation-noise-v1` schema make trajectories shard-independent; larger candidate sizes share the first N seeds. Multi-trajectory policies decode and composite each reconstruction, aggregate them voxel-wise by mean or median, then composite again. This aggregates trajectories from one checkpoint, not separately trained models.

4 Experiments

4.1 Data and Split Protocol

The ASNR-MICCAI BraTS Local Synthesis release [10,13,4,1,2,3] contains 1,251 training cases with complete and voided T1n, a full mask, and healthy and unhealthy sub-masks. Organizer-validation ground truth is hidden.

A deterministic 100-case hold-out (seed 2026) leaves 1,151 optimization cases; hold-out IDs are excluded from both the loader and component bank. Its checked-in 25-case development subset selects inference policy; the complementary 75-case manifest is used by all confirmation pipelines. Development and confirmation patients are disjoint, but the split is case-keyed: 3/25 development and

11/75 confirmation cases have an optimization-pool sibling timepoint. The five diagnostic cases have none. We add a post-hoc sensitivity analysis on the 64 confirmation cases patient-disjoint from optimization.

4.2 Training and Checkpoint Selection

We compare three max-normalized, 17-channel concatenation training variants with fixed-mask, random-augmentation, or weighted-mixture healthy-mask policies. Each uses training seed 0, batch size one per GPU, AdamW [11] with learning rate 10^{-5} and zero weight decay, a 300,000-step linear-decay horizon, and EMA decay 0.9999. Training is FP32 on NVIDIA A100 40 GB GPUs; checkpoints are saved every 5,000 steps.

Every 10,000 steps, EMA diagnostics use one deterministic full 1000-step DDPM trajectory on five development cases (fixed-mask 200,000 is absent). Within each variant, recorded checkpoints are ranked by aggregate structural similarity index measure (SSIM, descending), peak signal-to-noise ratio (PSNR, descending), and mean squared error (MSE, ascending), then by their mean rank. Fixed-mask 170,000 and 180,000 tie; we retain 180,000 because it leads PSNR and MSE. Random 290,000 and weighted-mixture 200,000 uniquely minimize joint rank. These diagnostics select checkpoints, not final performance.

4.3 Inference Selection and Evaluation

At each terminal EMA checkpoint (step 299,999), the 25 development cases compare nested $N \in \{1, 3, 5\}$ trajectories with voxel-wise mean or median aggregation. Global seed 0 and `brats-validation-noise-v1` give stable case and member seeds and shared first members across N . Mean $N = 5$ is selected for random and weighted mixture. For fixed mask, mean $N = 3$ lowers joint rank from 2.9200 to 2.7867 at three times the trajectories, while mean $N = 5$ is worse (3.2400); we retain $N = 1$. Policies transfer unchanged to selected checkpoints and are frozen before confirmation.

The official evaluator normalizes prediction and target to $[0, 1]$ from the voided-image 0.5th and 99.5th percentiles and scores only the provided healthy mask. We report SSIM and PSNR (higher is better), MSE (lower is better), and their mean within-case rank among selected pipelines. Means and sample SDs describe 75-case heterogeneity, not training-seed variability. Per metric, a Friedman test precedes two-sided paired Wilcoxon tests, Holm-adjusted over three contrasts. Mean paired improvements use deterministic 100,000-resample case-bootstrap 95% intervals (seed 2026). Compressed NIfTI predictions, per-case metrics, split audits, hashes, and metadata are retained as CSV/JSON. Code, paper artifacts, and analysis scripts will be released at <https://github.com/simonwinther/brats2026>.

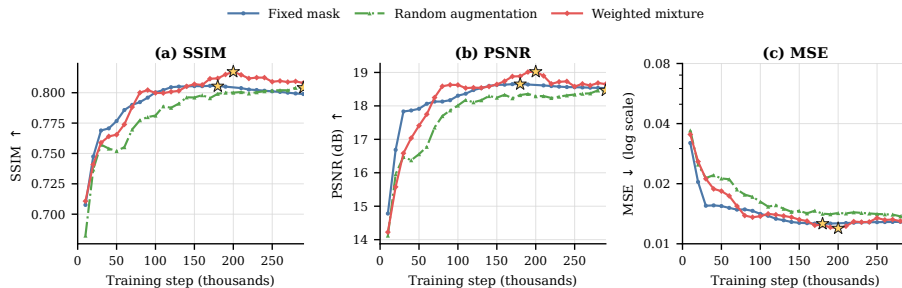


Fig. 3. Raw five-case EMA checkpoint diagnostics without dashboard smoothing. Stars mark the retained checkpoint after ranking aggregate SSIM, PSNR, and MSE; fixed-mask 170k and 180k tie in average rank, with 180k retained because it leads PSNR and MSE. Each point uses one deterministic 1000-step trajectory. Curves are selection diagnostics, not confirmation results.

Table 1. Development-set joint rank at terminal EMA checkpoints (lower is better). Bold and underlining mark the lowest and second-lowest value in each numeric column; fixed-mask $N = 1$ was retained as explained in the text.

Configuration	Fixed mask	Random aug.	Weighted mixture
$N = 1$	<u>2.9200</u>	4.7067	4.4000
Mean $N = 3$	2.7867	<u>2.1333</u>	2.6933
Median $N = 3$	2.9733	3.5333	3.7733
Mean $N = 5$	3.2400	1.6000	1.5867
Median $N = 5$	3.0800	3.0267	<u>2.5467</u>

5 Results

5.1 Development Selection

Figure 3 shows the periodic five-case development diagnostics and selected EMA checkpoints. Fixed-mask steps 170,000 and 180,000 tie for the lowest joint rank; step 180,000 is retained because it has the best aggregate PSNR and MSE (and third-best SSIM). Random step 290,000 and weighted-mixture step 200,000 are best on all three aggregate metrics within their respective recorded histories.

Table 1 summarizes inference-policy selection at the terminal EMA checkpoints on the 25-case development set, including the five diagnostic cases. Mean $N = 5$ has the lowest joint rank for random and weighted-mixture augmentation. The fixed-mask variant’s nominal optimum is mean $N = 3$, but its rank advantage over $N = 1$ is 0.1333 for three times as many reverse trajectories; mean $N = 5$ is additionally worse. We therefore select fixed mask $N = 1$ and random/weighted-mixture mean $N = 5$.

Table 2. Held-out 75-case confirmation results. Values are mean $\pm$ sample SD. Joint rank averages each pipeline’s per-case SSIM, PSNR, and MSE ranks; lower is better. Bold and underlining mark the best and second-best value in each numeric column. Fixed, Random, and Weighted denote the fixed-mask, random-augmentation, and weighted-mixture pipelines. Fixed uses $N = 1$, so its contrasts with the mean- $N = 5$ pipelines are not compute matched.

Pipeline	SSIM $\uparrow$	PSNR (dB) $\uparrow$	MSE $\downarrow$ ($\times 10^{-2}$)	Rank $\downarrow$
Fixed ($N = 1$)	0.7561 ± 0.1615	17.84 ± 1.75	1.3645 ± 0.8066	2.551
Random (mean $N = 5$)	0.7784 ± 0.1398	18.24 ± 2.19	1.2515 ± 0.7176	2.076
Weighted (mean $N = 5$)	0.7972 ± 0.1308	19.18 ± 1.80	0.9874 ± 0.5359	1.373

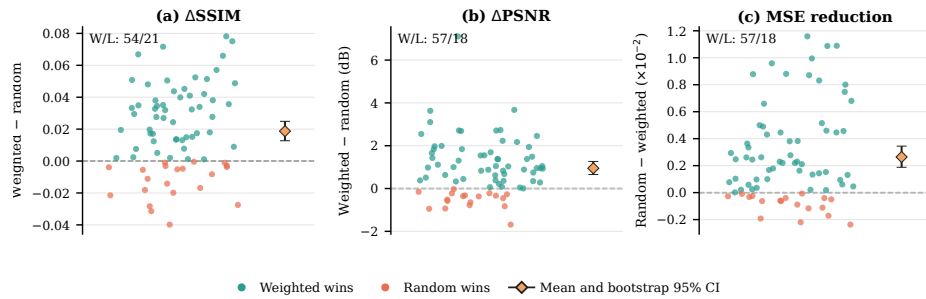


Fig. 4. Compute-matched paired effects of weighted-mixture versus random augmentation on all 75 confirmation cases. Positive values favor weighted mixture; each point is one case, and diamonds show the mean with the deterministic 100,000-resample case-bootstrap 95% interval. W/L gives weighted wins/losses (no ties). MSE is plotted as a reduction, so positive is better. Both pipelines use the same five case-stable trajectories and voxel-wise mean aggregation.

5.2 Held-out 75-Case Confirmation

Table 2 summarizes the held-out confirmation results.

The omnibus difference is significant for SSIM ($\chi_F^2(2) = 52.83$), PSNR, and MSE (both $\chi_F^2(2) = 52.91$; all $p < 3.4 \times 10^{-12}$). In the non-compute-matched comparison with fixed mask, weighted mixture improves mean SSIM by 0.0411 (95% CI: 0.0324 to 0.0500), PSNR by 1.34 dB (1.08 to 1.61), and reduces raw $[0, 1]$ -scale MSE by 0.003771 (0.002882 to 0.004746).

Relative to the compute-matched random-augmentation pipeline, weighted mixture improves SSIM by 0.0188 (0.0127 to 0.0249), PSNR by 0.948 dB (0.656 to 1.265), and reduces raw $[0, 1]$ -scale MSE by 0.002641 (0.001878 to 0.003447). All nine metric-specific pairwise tests remain significant after Holm correction within each metric family (largest adjusted $p = 0.0178$).

Figure 4 shows that the compute-matched gain is not universal but occurs in most cases: weighted mixture wins 54/75 cases on SSIM and 57/75 on both PSNR and MSE. It improves all three metrics against random augmentation in

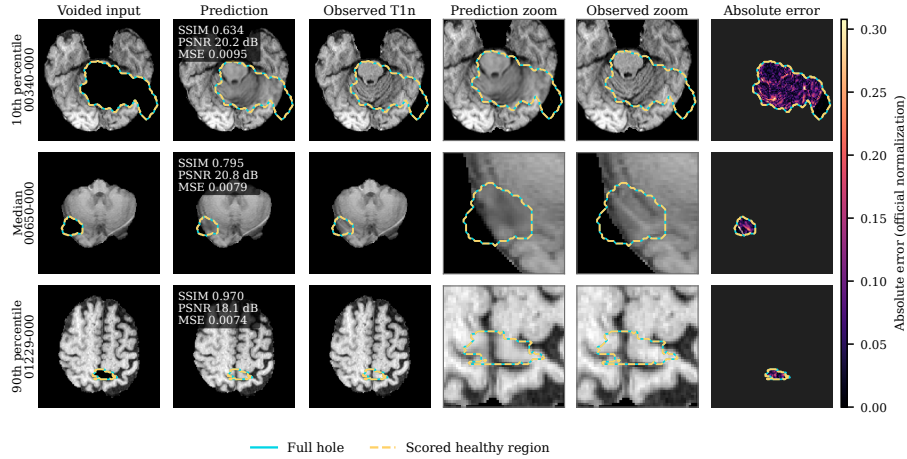


Fig. 5. Weighted-mixture mean- $N = 5$ cases at empirical SSIM quantiles 10/50/90. Columns show voided input, prediction, observed T1n, paired zooms, and normalized absolute error. Cyan denotes the full hole and dashed yellow the scored healthy region; error is shown there on a shared clipped scale. Axial slices maximize mask coverage independently of reconstruction error.

51 cases, loses all three in 15, and has mixed directions in 9. Among all three selected pipelines, weighted mixture is simultaneously best on all three metrics in 46 cases and obtains the lowest mean joint rank.

Figure 5 illustrates substantial variation in case difficulty. The low-SSIM example retains gross structure but has a localized intensity and boundary mismatch; the median is coherent with residual error in the scored region; and the high-SSIM example closely matches the target there. Because the observed T1n may be pathological elsewhere in the hole, its score is not evidence of counterfactual accuracy over the full cyan outline.

6 Discussion

Weighted mixture gives the strongest selected pipeline on held-out data. Random and weighted use the same architecture, training horizon, mean- $N = 5$ inference, split, seeds, and evaluator; their contrast therefore matches inference compute. Their training policies differ jointly in source families, size sampling, transforms, whether current-case sources are eligible, epoch refresh, and fallback. Evidence supports the full policy, not its configured 80/15/5 proportions alone. Gains in most paired cases argue against an outlier-only effect, but 15 all-metric losses show it is not uniformly superior. One training seed and arm-specific checkpoints prevent component-level causal attribution.

Fixed mask uses $N = 1$: mean $N = 3$ improved development joint rank by only 0.1333 at triple trajectory cost, while mean $N = 5$ was worse. Thus only weighted versus random is trajectory-matched; fixed versus augmented is not.

Invertible wavelets enable full-volume 3D sampling and hard compositing preserves observed tissue, but mean $N = 5$ needs five 1000-step ancestral trajectories. Faster wavelet inpainting [5] motivates validation under the same frozen cohort and scoring protocol.

Metrics cover artificially removed healthy voxels. Because observed T1n remains pathological in m_u , neither the metrics nor the qualitative reference establish patient-specific counterfactual correctness there; such ground truth does not exist. Plausible reconstructions may hallucinate incorrect anatomy and are not clinically validated. Hidden organizer-validation ground truth prevents local external scoring.

The case-keyed split is not fully patient-independent: optimization includes sibling timepoints for 3/25 development and 11/75 confirmation cases; development and confirmation patients are mutually disjoint. In a post-hoc 64-case patient-disjoint analysis, weighted mixture obtains SSIM 0.8101, PSNR 19.27 dB, and MSE 0.009220; random obtains 0.7911, 18.33 dB, and 0.011664; and fixed obtains 0.7703, 17.93 dB, and 0.012764. The ordering is unchanged. This removes confirmation overlap only, not possible influence on development-stage selection.

Other limitations are one seed per variant, five-case one-trajectory checkpoint diagnostics, transfer of terminally selected inference policies to arm-specific checkpoints, and no completed external baseline. Across-case variability does not measure seed variability, so prior-method superiority is unestablished. Development ranks lack their original per-case predictions; one random-arm dirty-worktree diff was not preserved, although its checkpoint, configuration, metrics, sampling metadata, and prediction hashes are archived. Future work should add seeds and baselines, preserve all outputs, validate faster sampling, and use patient-level splits and multimodal conditioning.

7 Conclusion

CATCH combines conditional 3D Haar-wavelet diffusion, tumor-excluded supervision, hard compositing, and development-selected aggregation. In this single-seed internal study, weighted-mixture augmentation is strongest under matched compute on artificially removed healthy tissue; clinical and cross-method superiority remain unestablished.

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